Lesson Planning

Complete Lesson Plan Book

Class: Class 11 – FSc Pre-Engineering

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Number Systems

Learning Objectives

• Define and explain real numbers in the context of Number Systems.
• Differentiate between complex numbers and binary operations with suitable examples.
• Solve numerical problems related to properties of numbers accurately.
• Apply the ideas of Number Systems to everyday situations involving number line.

Key Concepts

• Meaning, definition and scope of real numbers.
• Important properties, types and examples of complex numbers.
• Relationship between binary operations and properties of numbers.
• Practical applications of Number Systems, especially number line.
• Key terms and a summary of the main points of Number Systems.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Number Systems with a daily-life example.
2. Explain real numbers on the board with definitions and clear diagrams.
3. Discuss complex numbers and binary operations using simple, relatable examples.
4. Involve the students through short questions on properties of numbers.
5. Solve a numerical on number line together with the class.
6. Summarize the lesson and highlight the key points of Number Systems.

Classroom Activities

• Group discussion: students compare complex numbers with binary operations.
• Board activity: solve a question on properties of numbers in pairs.
• Quick quiz on the key terms of Number Systems.

Assessment Questions

1. Define real numbers.
2. Differentiate between complex numbers and binary operations.
3. Solve a numerical based on properties of numbers.
4. Write a short note on number line.

Homework

• Read Number Systems from the book and solve the exercise questions.
• Prepare short notes on real numbers and complex numbers.

Time Allocation

4–6 periods of 40 minutes each

Unit 2: Sets, Functions and Groups

Learning Objectives

• Define and explain set operations in the context of Sets, Functions and Groups.

• Differentiate between relations and functions with suitable examples.

• Solve numerical problems related to binary operations accurately.

• Apply the ideas of Sets, Functions and Groups to everyday situations involving group axioms.

Key Concepts

• Meaning, definition and scope of set operations.

• Important properties, types and examples of relations.

• Relationship between functions and binary operations.

• Practical applications of Sets, Functions and Groups, especially group axioms.

• Key terms and a summary of the main points of Sets, Functions and Groups.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Sets, Functions and Groups with a daily-life example.
2. Explain set operations on the board with definitions and clear diagrams.
3. Discuss relations and functions using simple, relatable examples.
4. Involve the students through short questions on binary operations.
5. Solve a numerical on group axioms together with the class.
6. Summarize the lesson and highlight the key points of Sets, Functions and Groups.

Classroom Activities

• Group discussion: students compare relations with functions.

• Board activity: solve a question on binary operations in pairs.

• Quick quiz on the key terms of Sets, Functions and Groups.

Assessment Questions

1. Define set operations.
2. Differentiate between relations and functions.
3. Solve a numerical based on binary operations.
4. Write a short note on group axioms.

Homework

• Read Sets, Functions and Groups from the book and solve the exercise questions.

• Prepare short notes on set operations and relations.

Time Allocation

4–6 periods of 40 minutes each

Unit 3: Matrices and Determinants

Learning Objectives

• Define and explain matrix operations in the context of Matrices and Determinants.

• Differentiate between inverse of matrix and determinants with suitable examples.

• Solve numerical problems related to Cramer's rule accurately.

• Apply the ideas of Matrices and Determinants to everyday situations involving system of equations.

Key Concepts

• Meaning, definition and scope of matrix operations.

• Important properties, types and examples of inverse of matrix.

• Relationship between determinants and Cramer's rule.

• Practical applications of Matrices and Determinants, especially system of equations.

• Key terms and a summary of the main points of Matrices and Determinants.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Matrices and Determinants with a daily-life example.
2. Explain matrix operations on the board with definitions and clear diagrams.
3. Discuss inverse of matrix and determinants using simple, relatable examples.
4. Involve the students through short questions on Cramer's rule.
5. Solve a numerical on system of equations together with the class.
6. Summarize the lesson and highlight the key points of Matrices and Determinants.

Classroom Activities

• Group discussion: students compare inverse of matrix with determinants.

• Board activity: solve a question on Cramer's rule in pairs.

• Quick quiz on the key terms of Matrices and Determinants.

Assessment Questions

1. Define matrix operations.
2. Differentiate between inverse of matrix and determinants.
3. Solve a numerical based on Cramer's rule.
4. Write a short note on system of equations.

Homework

• Read Matrices and Determinants from the book and solve the exercise questions.

• Prepare short notes on matrix operations and inverse of matrix.

Time Allocation

4–6 periods of 40 minutes each

Unit 4: Quadratic Equations

Learning Objectives

• Define and explain factorization in the context of Quadratic Equations.

• Differentiate between quadratic formula and discriminant with suitable examples.

• Solve numerical problems related to nature of roots accurately.

• Apply the ideas of Quadratic Equations to everyday situations involving sum and product of roots.

Key Concepts

• Meaning, definition and scope of factorization.

• Important properties, types and examples of quadratic formula.

• Relationship between discriminant and nature of roots.

• Practical applications of Quadratic Equations, especially sum and product of roots.

• Key terms and a summary of the main points of Quadratic Equations.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Quadratic Equations with a daily-life example.
2. Explain factorization on the board with definitions and clear diagrams.
3. Discuss quadratic formula and discriminant using simple, relatable examples.
4. Involve the students through short questions on nature of roots.
5. Solve a numerical on sum and product of roots together with the class.
6. Summarize the lesson and highlight the key points of Quadratic Equations.

Classroom Activities

• Group discussion: students compare quadratic formula with discriminant.

• Board activity: solve a question on nature of roots in pairs.

• Quick quiz on the key terms of Quadratic Equations.

Assessment Questions

1. Define factorization.
2. Differentiate between quadratic formula and discriminant.
3. Solve a numerical based on nature of roots.
4. Write a short note on sum and product of roots.

Homework

• Read Quadratic Equations from the book and solve the exercise questions.

• Prepare short notes on factorization and quadratic formula.

Time Allocation

4–6 periods of 40 minutes each

Unit 5: Partial Fractions

Learning Objectives

• Define and explain proper fractions in the context of Partial Fractions.
• Differentiate between decomposition and repeated factors with suitable examples.
• Solve numerical problems related to quadratic factors accurately.
• Apply the ideas of Partial Fractions to everyday situations involving applications.

Key Concepts

• Meaning, definition and scope of proper fractions.
• Important properties, types and examples of decomposition.
• Relationship between repeated factors and quadratic factors.
• Practical applications of Partial Fractions, especially applications.
• Key terms and a summary of the main points of Partial Fractions.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Partial Fractions with a daily-life example.
2. Explain proper fractions on the board with definitions and clear diagrams.
3. Discuss decomposition and repeated factors using simple, relatable examples.
4. Involve the students through short questions on quadratic factors.
5. Solve a numerical on applications together with the class.
6. Summarize the lesson and highlight the key points of Partial Fractions.

Classroom Activities

• Group discussion: students compare decomposition with repeated factors.
• Board activity: solve a question on quadratic factors in pairs.
• Quick quiz on the key terms of Partial Fractions.

Assessment Questions

1. Define proper fractions.
2. Differentiate between decomposition and repeated factors.
3. Solve a numerical based on quadratic factors.
4. Write a short note on applications.

Homework

• Read Partial Fractions from the book and solve the exercise questions.
• Prepare short notes on proper fractions and decomposition.

Time Allocation

4–6 periods of 40 minutes each

Unit 6: Sequences and Series

Learning Objectives

• Define and explain arithmetic progression in the context of Sequences and Series.

• Differentiate between geometric progression and harmonic progression with suitable examples.

• Solve numerical problems related to sum of series accurately.

• Apply the ideas of Sequences and Series to everyday situations involving means.

Key Concepts

• Meaning, definition and scope of arithmetic progression.

• Important properties, types and examples of geometric progression.

• Relationship between harmonic progression and sum of series.

• Practical applications of Sequences and Series, especially means.

• Key terms and a summary of the main points of Sequences and Series.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Sequences and Series with a daily-life example.
2. Explain arithmetic progression on the board with definitions and clear diagrams.
3. Discuss geometric progression and harmonic progression using simple, relatable examples.
4. Involve the students through short questions on sum of series.
5. Solve a numerical on means together with the class.
6. Summarize the lesson and highlight the key points of Sequences and Series.

Classroom Activities

• Group discussion: students compare geometric progression with harmonic progression.

• Board activity: solve a question on sum of series in pairs.

• Quick quiz on the key terms of Sequences and Series.

Assessment Questions

1. Define arithmetic progression.
2. Differentiate between geometric progression and harmonic progression.
3. Solve a numerical based on sum of series.
4. Write a short note on means.

Homework

• Read Sequences and Series from the book and solve the exercise questions.

• Prepare short notes on arithmetic progression and geometric progression.

Time Allocation

4–6 periods of 40 minutes each

Unit 7: Permutation, Combination and Probability

Learning Objectives

• Define and explain factorial notation in the context of Permutation, Combination and Probability.

• Differentiate between permutations and combinations with suitable examples.

• Solve numerical problems related to probability rules accurately.

• Apply the ideas of Permutation, Combination and Probability to everyday situations involving binomial distribution.

Key Concepts

• Meaning, definition and scope of factorial notation.

• Important properties, types and examples of permutations.

• Relationship between combinations and probability rules.

• Practical applications of Permutation, Combination and Probability, especially binomial distribution.

• Key terms and a summary of the main points of Permutation, Combination and Probability.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Permutation, Combination and Probability with a daily-life example.
2. Explain factorial notation on the board with definitions and clear diagrams.
3. Discuss permutations and combinations using simple, relatable examples.
4. Involve the students through short questions on probability rules.
5. Solve a numerical on binomial distribution together with the class.
6. Summarize the lesson and highlight the key points of Permutation, Combination and Probability.

Classroom Activities

• Group discussion: students compare permutations with combinations.

• Board activity: solve a question on probability rules in pairs.

• Quick quiz on the key terms of Permutation, Combination and Probability.

Assessment Questions

1. Define factorial notation.
2. Differentiate between permutations and combinations.
3. Solve a numerical based on probability rules.
4. Write a short note on binomial distribution.

Homework

• Read Permutation, Combination and Probability from the book and solve the exercise questions.

• Prepare short notes on factorial notation and permutations.

Time Allocation

4–6 periods of 40 minutes each

Unit 8: Mathematical Induction and Binomial Theorem

Learning Objectives

• Define and explain induction principle in the context of Mathematical Induction and Binomial Theorem.

• Differentiate between binomial expansion and general term with suitable examples.

• Solve numerical problems related to Pascal's triangle accurately.

• Apply the ideas of Mathematical Induction and Binomial Theorem to everyday situations involving applications.

Key Concepts

• Meaning, definition and scope of induction principle.

• Important properties, types and examples of binomial expansion.

• Relationship between general term and Pascal's triangle.

• Practical applications of Mathematical Induction and Binomial Theorem, especially applications.

• Key terms and a summary of the main points of Mathematical Induction and Binomial Theorem.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Mathematical Induction and Binomial Theorem with a daily-life example.

2. Explain induction principle on the board with definitions and clear diagrams.

3. Discuss binomial expansion and general term using simple, relatable examples.

4. Involve the students through short questions on Pascal's triangle.

5. Solve a numerical on applications together with the class.

6. Summarize the lesson and highlight the key points of Mathematical Induction and Binomial Theorem.

Classroom Activities

• Group discussion: students compare binomial expansion with general term.

• Board activity: solve a question on Pascal's triangle in pairs.

• Quick quiz on the key terms of Mathematical Induction and Binomial Theorem.

Assessment Questions

1. Define induction principle.

2. Differentiate between binomial expansion and general term.

3. Solve a numerical based on Pascal's triangle.

4. Write a short note on applications.

Homework

• Read Mathematical Induction and Binomial Theorem from the book and solve the exercise questions.

• Prepare short notes on induction principle and binomial expansion.

Time Allocation

4–6 periods of 40 minutes each

Unit 9: Fundamentals of Trigonometry

Learning Objectives

• Define and explain angle measurement in the context of Fundamentals of Trigonometry.
• Differentiate between trigonometric ratios and allied angles with suitable examples.
• Solve numerical problems related to graphs of functions accurately.
• Apply the ideas of Fundamentals of Trigonometry to everyday situations involving identities.

Key Concepts

• Meaning, definition and scope of angle measurement.
• Important properties, types and examples of trigonometric ratios.
• Relationship between allied angles and graphs of functions.
• Practical applications of Fundamentals of Trigonometry, especially identities.
• Key terms and a summary of the main points of Fundamentals of Trigonometry.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Fundamentals of Trigonometry with a daily-life example.
2. Explain angle measurement on the board with definitions and clear diagrams.
3. Discuss trigonometric ratios and allied angles using simple, relatable examples.
4. Involve the students through short questions on graphs of functions.
5. Solve a numerical on identities together with the class.
6. Summarize the lesson and highlight the key points of Fundamentals of Trigonometry.

Classroom Activities

• Group discussion: students compare trigonometric ratios with allied angles.
• Board activity: solve a question on graphs of functions in pairs.
• Quick quiz on the key terms of Fundamentals of Trigonometry.

Assessment Questions

1. Define angle measurement.
2. Differentiate between trigonometric ratios and allied angles.
3. Solve a numerical based on graphs of functions.
4. Write a short note on identities.

Homework

• Read Fundamentals of Trigonometry from the book and solve the exercise questions.
• Prepare short notes on angle measurement and trigonometric ratios.

Time Allocation

4–6 periods of 40 minutes each

Unit 10: Trigonometric Identities

Learning Objectives

• Define and explain sum and difference formulas in the context of Trigonometric Identities.
• Differentiate between double angle and half angle with suitable examples.
• Solve numerical problems related to solution of triangles accurately.
• Apply the ideas of Trigonometric Identities to everyday situations involving heights and distances.

Key Concepts

• Meaning, definition and scope of sum and difference formulas.
• Important properties, types and examples of double angle.
• Relationship between half angle and solution of triangles.
• Practical applications of Trigonometric Identities, especially heights and distances.
• Key terms and a summary of the main points of Trigonometric Identities.

Teaching Methodology (Step by Step)

1. Recap the previous lesson and introduce Trigonometric Identities with a daily-life example.
2. Explain sum and difference formulas on the board with definitions and clear diagrams.
3. Discuss double angle and half angle using simple, relatable examples.
4. Involve the students through short questions on solution of triangles.
5. Solve a numerical on heights and distances together with the class.
6. Summarize the lesson and highlight the key points of Trigonometric Identities.

Classroom Activities

• Group discussion: students compare double angle with half angle.
• Board activity: solve a question on solution of triangles in pairs.
• Quick quiz on the key terms of Trigonometric Identities.

Assessment Questions

1. Define sum and difference formulas.
2. Differentiate between double angle and half angle.
3. Solve a numerical based on solution of triangles.
4. Write a short note on heights and distances.

Homework

• Read Trigonometric Identities from the book and solve the exercise questions.
• Prepare short notes on sum and difference formulas and double angle.

Time Allocation

4–6 periods of 40 minutes each